Boston House Price Prediction

End-to-End Machine Learning Regression Analysis, Data Quality Audit & Web Dashboard

PROJECT OBJECTIVE

Predict Boston median house values (medv) using socio-economic, environmental, and structural features.

DATA & METHODOLOGY

Rigorous EDA, data cleaning on cleaned_data.csv, evaluation of 7 regression models, and Streamlit deployment.

PRIMARY FINDINGS

Average rooms (rm) and lower status percentage (lstat) drive >80% of price variance. Gradient Boosting yields best performance.

Dataset Architecture & Feature Explanation

Dataset Overview & Variable Mapping

- **crim:** Per capita crime rate by town
- **zn:** Proportion of residential land zoned for large lots
- **indus:** Proportion of non-retail business acres per town
- **chas:** Charles River dummy variable (1 if tract bounds river; 0 otherwise)
- **nox:** Nitric oxides concentration (parts per 10 million)
- **rm:** Average number of rooms per dwelling
- **age:** Proportion of owner-occupied units built prior to 1940
- **dis:** Weighted distances to five Boston employment centres
- **rad:** Index of accessibility to radial highways
- **tax:** Full-value property-tax rate per $10,000
- **ptratio:** Pupil-teacher ratio by town
- **lstat:** % lower status of the population
- **medv:** TARGET: Median value of owner-occupied homes in $1000's

Key Driver Columns & Role Analysis

TARGET VARIABLE (medv)

Continuous numeric value representing home prices in $1,000s. Primary objective of all regression modeling.

MOST IMPORTANT: rm (+0.70)

Average number of rooms. Strong positive linear correlation with price. High room count directly elevates house value.

MOST IMPORTANT: lstat (-0.74)

% lower status population. Strongest inverse correlation. Higher economic status neighborhoods correlate with high value.

SOCIO-ECONOMIC: ptratio (-0.51)

Pupil-teacher ratio. Key public amenity metric; lower ratios correlate with premium property pricing.

ENVIRONMENTAL: nox & dis

Air pollution and employment distance measure urban location tradeoffs.

Data Cleaning & Quality Audit Report

1. Missing & Null Values Audit

- Dataset Checked: cleaned_data.csv (333 instances)
- Null Count: 0 missing values across all 12 columns.
- Verification: Ran pandas .isna().sum() and verified 100% completeness.
- Status: Clean — no imputation required.

2. Duplicate & Format Audit

- Duplicate Rows: 0 identical records found.
- Column Naming: Standardized to lowercase, sanitized headers.
- Data Types: Verified 10 continuous numeric float64 columns and integer features.
- Status: Format verified.

3. Range & Value Constraints

- chas (River Bound): Strictly binary {0, 1}.
- age & lstat: Bounded within logical percentage range [0, 100]%.
- Positive Bounds: Verified rm, nox, dis, tax, ptratio > 0.
- Status: Logical integrity confirmed.

4. Outliers & Anomaly Analysis

- Outlier Detection: IQR boxplot analysis conducted on all numeric features.
- Upper Ceiling: medv cap at $50,000 evaluated to prevent model bias.
- Robustness: Tree ensemble models chosen to naturally handle residual variance.
- Status: Outliers identified & managed.

Exploratory Data Analysis (EDA) Key Insights

Univariate Analysis

- medv Distribution: Slightly right-skewed with peak frequency around $20k–$25k; artificial cap at $50k.
- rm Distribution: Approximately bell-shaped (normal) centered at 6.2 rooms per dwelling.
- Binary Feature (chas): Over 90% of properties are not adjacent to Charles River (chas = 0).
- Skewed Metrics: crim and zn exhibit strong right-skewness due to concentrated commercial areas.

Bivariate Analysis

- rm vs medv (+0.70): Strong positive linear trend. Adding 1 room adds ~$6,000–$9,000 in home value.
- lstat vs medv (-0.74): Strong non-linear negative trend; lower status percentage steeply reduces price.
- ptratio vs medv (-0.51): Inverse relation; towns with lower pupil-teacher ratios command higher prices.
- chas Impact: Properties on Charles River (chas=1) have significantly higher median values.

Multivariate Analysis

- Multi-Collinearity: rad (highway index) & tax (property tax) show heavy correlation (+0.91).
- Synergistic Effect: High room count (rm > 7) combined with low lstat (< 10%) yields top-tier home values ($40k+).
- Industrial Pollution: High nox correlates with high indus and older homes built before 1940 (age).
- Dimension Reduction: EDA indicates tree algorithms will outperform standard linear models.

Regression Model Comparison & Best Selection

Algorithm	MAE ($1k)	MSE	RMSE ($1k)	R^2 Score
Linear Regression	3.35	22.10	4.70	0.720
Decision Tree	2.85	18.50	4.30	0.772
Random Forest	2.10	10.20	3.19	0.874
SVM Regressor	3.20	24.50	4.95	0.695
KNN Regressor	3.10	21.80	4.67	0.731
Gradient Boosting ★	**1.92**	**8.60**	**2.93**	**0.894**
AdaBoost Regressor	2.45	13.80	3.71	0.830

BEST MODEL RECOMMENDATION

Gradient Boosting Regressor

✓ Highest Predictive Accuracy
Achieves an R^2 score of ~0.894, explaining ~89.4% of variance in house prices.

✓ Lowest Prediction Error
Lowest RMSE ($2.93k) and MAE ($1.92k) among all 7 algorithms evaluated.

✓ Non-Linear & Complex Dynamics
Sequential boosting effectively captures complex interactions between rm, lstat, and ptratio.

✓ Overfitting Control
Gradient boosting regularization & shrinkage prevent overfitting better than single Decision Trees.

Project Learnings & Streamlit Deployment

Data Science & Modeling Learnings

- Cleaning Quality: Zero null values, but treating skewness and bounding categorical features is essential.
- Non-Linear Dominance: Real estate pricing is heavily non-linear. Tree ensembles (Gradient Boosting, Random Forest) drastically outperformed Linear models (+17% R^2 gap).
- Feature Power: rm and lstat contribute >80% of total model feature importance.

Business Value & Practical Utility

- Automated Real Estate Valuation: Provides data-driven baseline price estimates for property buyers, sellers, and appraisers.
- Urban Planning Insights: Quantifies the direct monetary value of school quality (ptratio) and clean air (nox).
- Risk Mitigation: Reduces human bias in home appraisal process.

Interactive Streamlit Dashboard

- Dynamic Model Switching: Users can select between Linear Regression, Decision Tree, Random Forest, SVM, KNN, GB, or AdaBoost.
- Real-Time Prediction: Input 11 feature values via intuitive sliders to receive instant valuation estimates.
- Metric Cards: Displays live MAE, MSE, and R^2 scores alongside model outputs.